

Hima Shanthi Rayapati

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Education

- Arizona State University**, Data Science, Analytics and Engineering (Human-Centered Applications) 08/2026 – 12/2027
- Ira A. Fulton Schools of Engineering Graduate Scholarship (awarded for academic performance) | ASU AI Scholars Program, 2026–27 (year-long cohort with workshops on applied AI tools)
- Arizona State University**, BS in Psychology and Mathematics (Statistics) 01/2023 – 05/2026
- **Minor:** Philosophy | **GPA:** 4.0/4.0

Research Experience

- Student Researcher**, CARE Lab (Prof. Qiaoning Zhang), ASU 08/2026 – Present
- Attend lab meetings and discuss ongoing research on human-AI interaction.
 - Developing my own research project on AI and human trust and learning (in progress).
- Research Assistant**, Thinking Across Languages and Contexts Lab, ASU 08/2025 – Present
- Assist a graduate student with testing experiments built in jsPsych and JavaScript.
 - Help check whether data are recorded correctly and organized for analysis.
 - Perform exploratory data analysis and assist in interpreting experimental results under supervision.
- Research Assistant**, Perception, Action, Language & Meaning Lab, ASU 08/2025 – Present
- Run two-person collaborative task sessions (e.g., Taboo), observing and recording coordination, communication, and performance data using VR, motion-capture, and eye-tracking.
 - Work with the lab's time-series behavioral data, learning how to clean, organize, and explore it.
 - Read papers and explore possible methods (e.g., multilevel vector autoregression) to understand which approaches might fit the lab's data.
 - Contribute to discussions by sharing preliminary notes and observations from trial analyses.
- Volunteer Research Assistant**, Longitudinal & Behavioral Data Science Lab, ASU 08/2024 – 12/2024
- Participated in bi-weekly lab meetings, contributing to discussions and staying updated on research progress.
- UX Researcher (UXR Guild)**, NEXT Lab, ASU 08/2026 – Present
- Test games, digital twins, and interactive apps with users to find usability problems and collect feedback.
 - Find and document bugs and performance issues and share them with developers.
 - Work with developers on UI and accessibility improvements.
- Research Assistant**, Evolution, Ecology, and Social Behavior Lab, ASU 08/2023 – 11/2023
- Assisted with survey design, pilot testing, data coding, and literature reviews in social cognition research.
 - Participated in lab meetings and contributed to discussions on study design and interpretation.

Projects

- Trust in Advice: AI vs. Human Sources (Data Analysis & Written Report)** *Trust in AI.com/2025*
- Analyzed 30,000+ human-AI interactions to study effects of advice source.
 - Modeled decision changes before and after advice exposure.
 - Found advice correctness as primary driver, with modest AI-related effects.
- Cultural Differences in Trust: AI vs. Human Expert Decision-Making** *Research Proposal / 2025*
- Proposed cross-cultural study (Mexico vs. U.S.) on AI vs. human decision-making under conflict.
 - Outlined experimental design focusing on competence and legitimacy.
 - Formulated mixed-effects modeling approach for analysis.
 - Synthesized literature across cognition, decision-making, and human-AI interaction.

Analyzing the Impact of AI on Stack Overflow Participation (NLP & Data Analysis)

GitHub

- Analyzed large-scale Stack Overflow data using Python to study changes in user behavior after AI tool releases.
- Built time-series datasets and computed engagement metrics (answer rates, response time, retention) using Python.
- Performed statistical comparisons and visualized platform-level trends.
- Explored tag-based trends and basic NLP features to examine topic shifts.

Stochastic Dual-Timescale Model of Learning (ODE & SDE Modeling)

Final Project / 2025

- Developed differential equation model of fast and slow learning processes.
- Extended to stochastic framework to examine effects of noise.
- Verified theoretical properties via simulation.
- Linked model behavior to memory retention and learning variability.

Probabilistic Reward Learning Task using PsychoPy

Reward Learning.com/2025

- Designed probabilistic reward-learning task with informative vs. non-informative features.
- Examined learning under uncertainty and perceptual noise.
- Collected and analyzed participant data using Python.

Fairness in ML: Detecting Bias in LLM Answers (CSE 575, in progress)

- Building a classifier that flags biased answers from an LLM, using the BBQ bias benchmark dataset.
- Testing whether the classifier can pick the least biased answer out of several options the LLM gives.

Research Proposal: Mirror Neurons, Social Synchrony, and Collaborative Learning

Proposal.com/2025

- Proposed a research framework synthesizing human and animal literature on mirror neuron systems and social synchrony.
- Formulated hypotheses linking coordination mechanisms to collaborative learning outcomes.

Additional projects and full write-ups available at my portfolio.

Work Experience

Graduate Course Assistant (Statistical Methods), ASU – Tempe, AZ

01/2027 – Present

- Grade assignments and exams and give feedback to students.
- Hold office hours and answer student questions about course content.
- Work under the direction of a faculty supervisor.

Research Aide, ASU – Tempe, AZ

12/2024 – Present

- Review, categorize, and structure user-generated data to ensure quality and consistency.
- Assign relevant tags to comments for improved organization and searchability.
- Evaluate AI-generated tags against human annotation standards, analyzing human-AI agreement and error patterns across prompt versions to guide model refinement.
- Perform data cleaning tasks, identifying and correcting inconsistencies to maintain data accuracy.
- Work with databases to structure data and ensure accessibility for future use.

Lead Survey Assistant, ASU – Tempe, AZ

05/2023 – Present

- Conduct phone-based survey interviews with currently employed and recently graduated ASU students, ensuring accurate and comprehensive data collection by clarifying responses and addressing any ambiguities.
- Assist in the hiring process, including interviewing and evaluating candidates.
- Perform data cleaning and ensure the integrity of datasets for analysis.
- Train new student workers in survey procedures and oversee testing and calibration of survey tools.

Intern, Public Safety Crisis Solution – Phoenix, AZ

08/2024 – 11/2024

- Performed quantitative analysis to support research, program development, and accreditation documentation.
- Contributed research-based content on clinical procedures addressing first responders' mental health initiatives.

Skills

- **Programming & Statistical Tools:** R, SAS, MATLAB, Python, JMP, JASP, SQL (Microsoft Access), Java, C, Bash, REST APIs
- **Machine Learning & NLP (Foundational):** Principal Component Analysis (PCA), Neural network basics, Supervised vs. Unsupervised learning, Model evaluation concepts, Introductory text analysis (Voyant Tools)
- **Experimental & Research Tools:** jsPsych, PsychoPy, JavaScript, HTML, CSS, Qualtrics, SurveyMonkey
- **Data Analysis & Visualization:** Tableau, Excel, Google Sheets, Alteryx
- **Design & Prototyping:** Figma, Figma Motion
- **Collaboration & Productivity:** Slack, Teams, Zoom, Word, Outlook, Adobe PowerPoint, OneDrive, Dropbox